# HLB-D2 · F · age not recorded · position D2 · tether arm C

neurosurveillance-eeg · HLB-fem-profsurgery-troubleshooting

allego_1__uid0806-16-57-38

post-surgery day not recorded · day 7 post-tether · body weight not recorded · 23.66 h · ZT 8.0 to 7.6 · criterion HUM · QC epoch 10 s

## LIFE TIMELINE




## VERDICT LADDER

**✓ 0 FILE** — PASS

*Is the recording file physically sound?*

**85,172,608 samples, 23.66 h**

vs data samples == timestamp samples, remainder 0

85,172,608 data samples = 85,172,608 timestamp samples, remainder 0 and 0

**✓ 1 IDENTITY** — PASS

*Which native channels belong to this animal?*

**EEG D/33 EMG D/34**

vs layout derived from the file, never assumed

2 channels resolved from sensors_by_port; 112 wire-free inputs available as an in-file floor

**✓ 2 CONNECTION** — PASS

*Is the electrode connected, judged against wire-free inputs in the same file?*

**86.5% usable, longest unbroken block 87.3 min**

vs 60 Hz < 415.0 uV (p1 of 112 wire-free inputs in this file), rms 10-150 uV, zero railed samples

7,371 of 8,517 epochs usable on every role, with an unbroken block of 87.3 min

**✓ 3 SCOREABILITY** — PASS

*Can the sleep question even be asked?*

**23.28 h in 9 block(s), r@1min 0.499**

vs >= 15 min inside blocks of >= 124 s, within-block r@1min > 0.30, wake 15-90%

longest block 576.4 min; persistence 0.499 against a shuffled -0.001

**◐ 4 ARCHITECTURE** — PROVISIONAL

*What did the animal do?*

**wake 43.3%, NREM 49.9%, asleep in light 70.4%**

vs no pass mark: this rung reports, it does not judge

the light-phase figure swings 66.0 points across a +/-25% threshold sweep; no human has scored any epoch, so accuracy is unmeasured; ...

**✓ 5 EXPORT** — PASS

*Can a human check us?*

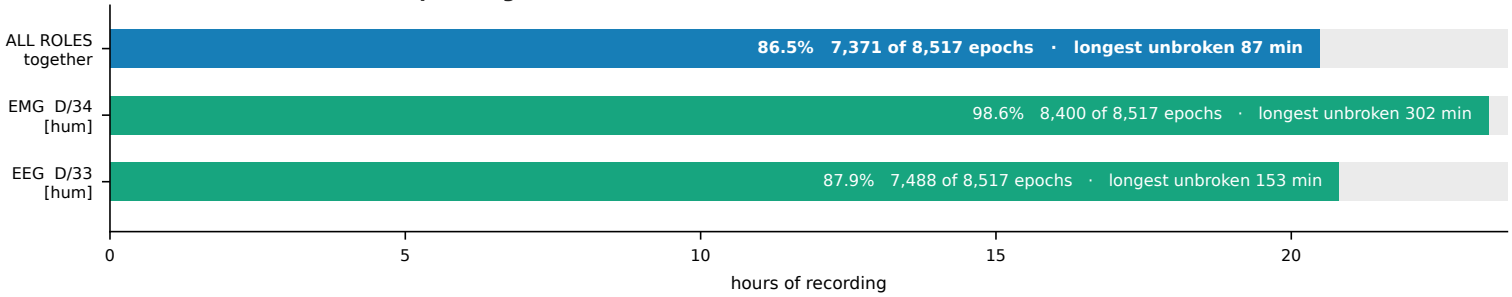
**4/4 pipeline artifacts, plain text written**

vs EDF for a scorer, plain text for the professor, both present

EDF and plain-text export both present

A rung is not evaluated if the rung below it failed: it reads BLOCKED. ? INSUFFICIENT means the question could not be asked, which is NOT the same as ✗ FAIL.

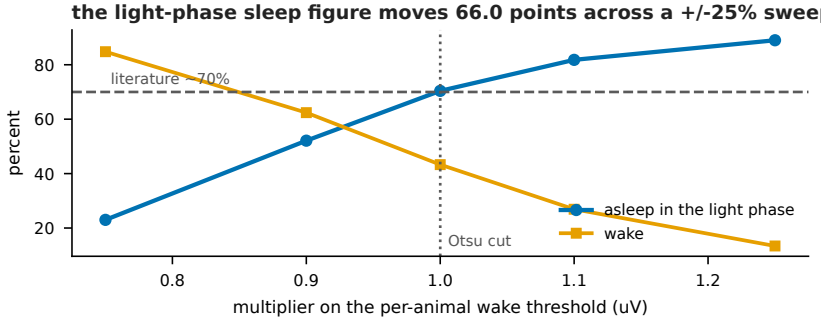
## RUNG 2 evidence — usable epochs against the in-file wire-free floor




falsification: the same criterion scores the 112 wire-free inputs in this file at 0.00% usable, and must be near zero · floor 60 Hz p1 415 uV, 1-40 Hz slope p1 -1.35

## RUNG 4 evidence — how much the wake threshold moves the answer

**the light-phase sleep figure moves 66.0 points across a +/-25% sweep of that one threshold**




**READ WITH THE PANEL ON THE LEFT**

- PROVISIONAL until a blind-scoring Cohen kappa exists. No human has scored any epoch from this project, so these labels have no known accuracy.
- The light-phase sleep figure moves 66.0 points across a +/-25% sweep of the wake threshold, so it is NOT threshold-independent and agreement with the literature is not independent confirmation.
- NO REM PERCENTAGE IS REPORTED: the percentile rule and an independent Otsu cut disagree by roughly a factor of five on this montage.

THE DEMONSTRATION THAT THE TETHER IS THE FIX. Went from 0.0 percent to 87.9 percent usable EEG after an arm C swap and nothing else: hum 916.9 to 133.2 uV, spectral slope -0.38 to -2.05, railing 29.7 to 0.0 percent, with its implant untouched. Tethered at day 8 after surgery, which is why long post-operative recovery is not required for a usable recording. ⚠A 40-sweep impedance block taken just after the swap said the swap had HARMED this animal, p = 8.4e-08. The block spanned 21 minutes and caught one long open state. The recording reversed it completely. This is the origin of the rule that impedance blocks screen and only a multi-hour recording is an

HLB-fem-profsurgery-troubleshooting · card schema v2 · generated 2026-08-09T11:11:53+09:00 · git not a git repo · thresholds c635d5245a5b (qc_thresholds.py) · criterion HUM at 10 s epochs. SLOPE and HUM usable fractions are NOT comparable, and neither is comparable to a fraction computed at the 4 s staging epoch.